

1. The question is about the interleaved execution of the two threads (Payroll and ATM withdrawal) presented in the lecture. However, in this case, the payroll thread starts first. If the initial balance B is \$200, what will be the final value of B after the interleaved execution of both threads.

B = \$200

• Interleaved Execution

- Load Rg, B
- add Rg, 1000

Interrupt

- Load Rg, B
- sub Rg, 80
- Store B, Rg

Return from int..

- Store B, Rg

1,200

2. Based on the textbook: a situation where several threads access and manipulate the same data concurrently while the outcome of the execution depends on the particular order in which the access takes place, is called a _____.

race
condition

3. Without inter-process communication mechanisms, these entities can share data:

Interrupt routine handler and main loop
threads within a multithreaded process

4. The shared-data problem or (race condition) arises when:

Threads access and manipulate shared data
Multiple threads execute concurrently

5. In order to address the data-shared problem, we can use

Mutual exclusion

6. Multiple threads of execution can occur when:

interrupts occur
multiple processes are concurrently executing
a process has multiple threads

7. True or False.

This question is related to the example using a payroll thread and an ATM withdrawal thread. If the two threads execute concurrently in an interleaved way (i.e., instructions from each would execute alternatively), the integrity of the shared variable B (balance) may get violated.

True

8. True or False.

This question is related to the example using a payroll thread and an ATM withdrawal thread. If

the two threads fully execute sequentially (i.e., one thread fully executes followed by the execution of the other thread), the integrity of the shared variable B (balance) may get violated.

False